Lecture 4: Compliment, Odds in Favour/Against, De Morgan's Law

29 March 2026 16:57

Topics of This Lecture:

1. Properties
2. Complement of Events
3. De Morgans Law
4. Subset
5. Axioms of Probability
6. Odds in Favour
   Odds in Against
7. Problems

1) Union and Intersection are :

1. Commutative:
- A U B = B U A
- A ∩ B = B ∩ A

2. Associative:
- A U (B U C) = (A U B) U C
- A ∩ (B ∩ C) = (A ∩ B) ∩ C

3. Distributive:
- (A U B) ∩ C = (A ∩ C) U (B ∩ C)
- (A ∩ B) U C = (A U C) ∩ (B U C)

2) Compliments of Events ($\bar{E}$):

$E \subseteq S$

$\bar{E} = 1 - E$

$\bar{E}$ E^c

$\bar{E} = S - E$

$P(\bar{H}) = ?$

$P(S) = 1$

Complement of events consists of all elements that are present in sample space (S) but not present within event (E).

$\bar{E}$ occurs if E doesn't occur

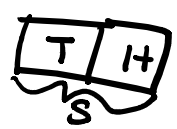
$\bar{E}$ = S - E

(|$\bar{E}$|/|S|) = (|S|/|S|) - (|E|/|S|)

P($\bar{E}$) = 1 - P(E)

$P(\bar{H}) = S - H$
$= 1 - H$

$P(\bar{H}) = P(S) - P(H)$
$= 1 - \frac{1}{2} = \frac{1}{2}$



$S - H = \bar{H}$
$S - T = \bar{T}$

3) De Morgan's Law:

- **$\overline{A \cup B} = \bar{A} \cap \bar{B}$**
- **$\overline{A \cap B} = \bar{A} \cup \bar{B}$**

4) Subset:

Let we have two events E & F.

If all outcomes in E are also present in F
Then E is contained in F

We can say E is subset of F

E ⊆ F

5) Axioms of Probability:

- **Axiom 1:**
  0 ≤ P(E) ≤ 1
- **Axiom 2:**
  P(S) = 1
- **Axiom 3:**
  For any sequence of mutually exclusive events E1, E2, E3,..... En (Ei ∩ Ej = 0 for all i,j)

  P(E1 U E2 U … U En) = P(E1) + P(E2) + … + P(En)

$$P\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} P(E_i)$$

Q. A total of 40% Indian use Gpay.
10 % use PhonePay
5 % use both
What % of Indian neither use Gpay nor PhonePay?

✓ Union
✓ Intersection
✓ Complements of Event

$P(G) = 0.4$

$P(P) = 0.1$

$P(G \cap P) = 0.05$

x G

x P

$P(G \cup P) = |G| + |P| - |G \cap P|$

$= 0.4 + 0.1 - 0.05$

$= 0.5 - 0.05 = 0.45$

$P(\overline{G \cup P}) = ?$

$\boxed{\overline{E} = S - E}$

$P(\overline{G \cup P}) = 1 - 0.45$

$= 0.55$

55% population

x Gpay

x Ppay.

6) Odds in Favour:

Ratio of no. of favourable outcomes to no. of unfavourable outcomes of a particular event is called odds in favour of an event.

Odds in favour of event **A = P(A)/P($\overline{A}$) = P(A)/1-P(A) = |A|/|$\overline{A}$|**

Odds in Against:

Ratio of no. of unfavourable outcomes to no. of favourable outcomes of a particular event is called odds in against of an event.

Odds in against of event **A = P($\overline{A}$)/P(A) = 1-P(A)/P(A) = |$\overline{A}$|/|A|**

Q2. Find the odds in favour & odds in against of throwing a die and getting a 6?

E = getting a 6

$\bar{E} = |S| - |E|$

$S = \{1, 2, 3, 4, 5, 6\}$

$|S| = 6$

$|E| = 1$

$|\bar{E}| = 6 - 1 = 5$

$|E| = 1$

$|\bar{E}| = 5$

$P(E) = \frac{|E|}{|S|} = \frac{1}{6}$

$P(\bar{E}) = \frac{|\bar{E}|}{|S|} = \frac{5}{6}$

odds in favour of $E = \frac{|E|}{|\bar{E}|} = \frac{1}{5}$

odds in against of $E = \frac{|\bar{E}|}{|E|} = \frac{5}{1}$

Q3. Odds in favour of India winning the world cup is 4:3
And odds in against of New Zealand winning is 2:6
Find the probability of :

- **India winning the cup**
- **New Zealand winning the cup**

I

N

odds in favour of $I = \frac{|I|}{|\bar{I}|} = \frac{4}{3}$

odds in against of $N = \frac{|\bar{N}|}{|N|} = \frac{2}{6}$

odds in favour of $I = \frac{|I|}{|\bar{I}|} = \frac{P(I)}{1 - P(I)} = \frac{4}{3}$

$\Rightarrow 3P(I) = 4 - 4P(I)$

$\Rightarrow 7P(I) = 4$

$$\Rightarrow 3P(I) = 4 - 4P(I)$$
$$\Rightarrow 7P(I) = 4$$
$$\Rightarrow P(I) = \frac{4}{7}$$

$$\text{odds in against of } N = \frac{|\bar{N}|}{|N|} = \frac{1 - P(N)}{P(N)} = \frac{2}{6}$$

$$\Rightarrow 6 - 6P(N) = 2P(N)$$
$$\Rightarrow 4P(N) = 6$$
$$\Rightarrow P(N) = \frac{6}{4} = \frac{3}{2}$$